

Equity-Aware Ambulance Outpost Siting in Sariyer

INDR 486/586 Term Project

June 8, 2026

Problem and Motivation

- Goal: locate ambulance outposts in Sarıyer at the district scale.
- Demand points: selected neighborhoods weighted by population.
- Candidate locations: public or quasi-public facilities.
- Main trade-off: fewer outposts vs. shorter and fairer access distances.
- Equity concern: avoiding solutions where some neighborhoods remain poorly served.

- 10 demand neighborhoods in Sarıyer.
- 7 candidate ambulance outpost locations.
- Population is used as the demand weight.
- Driving-distance matrix is used as the accessibility input.
- Reproducible workflow: CSV data → optimization model → frontier, assignments, and sensitivity outputs.

Optimization Model

Decision variables:

$$y_j = \begin{cases} 1 & \text{if facility } j \text{ is opened} \\ 0 & \text{otherwise} \end{cases} \quad x_{ij} = \begin{cases} 1 & \text{if demand } i \text{ is assigned to } j \\ 0 & \text{otherwise} \end{cases}$$

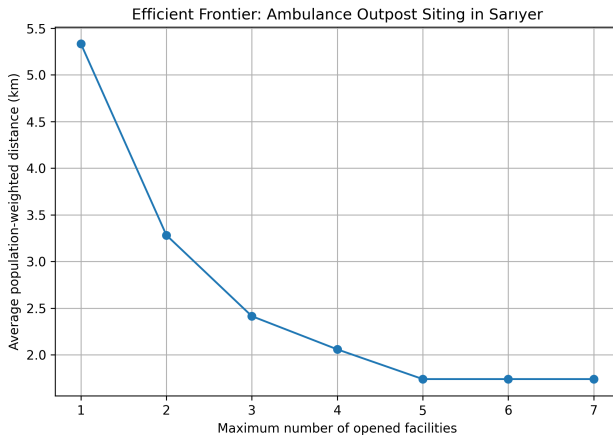
Objective:

$$\min \frac{\sum_{i \in I} \sum_{j \in J} w_i d_{ij} x_{ij}}{\sum_{i \in I} w_i}$$

Main constraints:

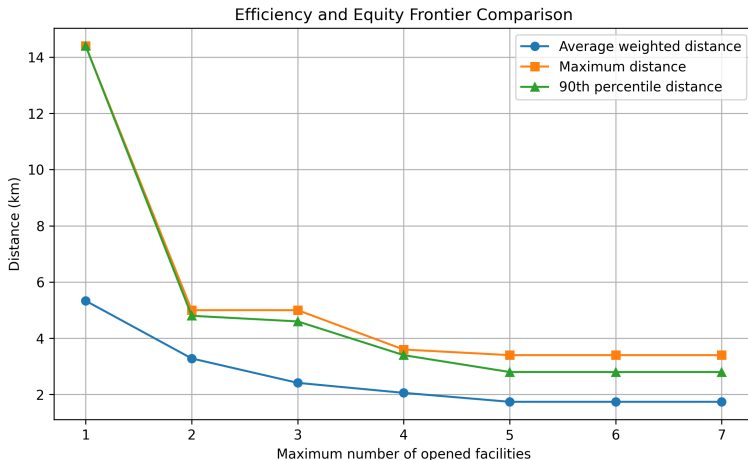
$$\sum_{j \in J} x_{ij} = 1, \quad x_{ij} \leq y_j, \quad \sum_{j \in J} y_j \leq \text{MaxFac}$$

Efficient Frontier



Average weighted distance decreases as more outposts are opened, but the marginal improvement becomes smaller after around four facilities.

Equity and Sensitivity Results

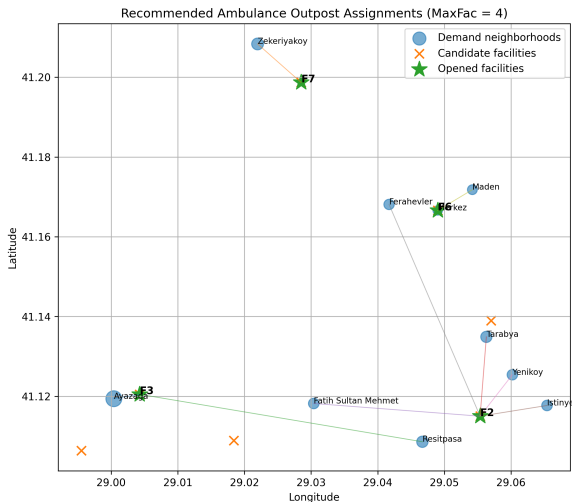


The comparison of average, maximum, and 90th percentile distances shows that four facilities substantially improves worst-case access.

Recommended Solution

- Recommended number of outposts: **4**.
- Opened facilities: **F2, F3, F6, F7**.
- Average weighted distance: approximately **2.06 km**.
- Maximum assigned distance: approximately **3.6 km**.
- Interpretation: four facilities provide a strong balance between efficiency, equity, and implementation cost.

Recommended Assignment Map



The selected facilities create local service clusters and reduce long-distance assignments, especially for the northern demand point.

Limitations and Conclusion

- Population is used as a proxy for EMS demand.
- Real EMS planning would require call-level demand and time-dependent travel times.
- The model does not include ambulance fleet size, queueing, or dispatch rules.
- Still, the framework is transparent, reproducible, and suitable for district-scale planning.
- Final recommendation: **open F2, F3, F6, and F7.**